INCH-POUND
MIL-PRF-7032/1
10 September 2010
SUPERSEDING
MS17406C
14 June 1974

PERFORMANCE SPECIFICATION SHEET

INVERTER, AIRCRAFT, DC TO AC, 750 VA 3-PHASE, 1000 VA 1-PHASE, AND 1000 VA 3-PHASE, MULTI-PURPOSE, CLASS B

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-7032.

SCOPE

1. Inverter class. The inverter is classified as a class B as modified herein.

2. Part or identifying number (PIN). Inverters covered by this specification sheet are identified using the following PIN:

Rotary inverter – M7032/1-1, M7032/1-2 (see figure 1)
Static inverter – M7032/1-3, M7032/1-4, M7032/1-5 (see figure 2)
Static inverter – M7032/1-6 (see figure 2)

APPLICABLE DOCUMENTS

1. Add the following specification to MIL-PRF-7032.

2.2.1 Specifications and standards. Add the following document:

MIL-M-7793/6	Meter, Time Totalizing, Miniature, Digital, 115 V 400 Hz. (Inactive for new design)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:48Z
Check the source to verify that this is the current version before use.

2.3 Non-Government publications. Add the following documents:

SAE-AS31061	Connector, Plug, Electric, Straight, Solder Contacts, AN Type. (DoD adopted)
SAE-AS31081	Connectors, Plug, Electric, Solder Contact, 90 Degree, AN Type. (DoD adopted)

REQUIREMENTS

1. Description. Power ratings and characteristics of the inverters covered by this specification sheet are shown in table I.

TABLE I. Inverter description.

Part no.	Power Rating 1/	Power factor	Max weight (lb)	Superseding
M7032/1-1	750VA, 3/1Φ	0.8 MIN LAG 0.95 MIN LEAD	23	MS17406-1A
M7032/1-2	750VA, 3/1Φ		23	MS17406-2A
M7032/1-3	750VA, 3Φ	0.75 MIN LAG 0.95 MIN LEAD	23	MS17406-3
M7032/1-4	1000VA, 1Φ		23	MS17406-4
M7032/1-5	1000VA, 1Φ, Low Input Voltage		23	MS17406-5
M7032/1-6	1000VA, 3Φ		23	

1/ Φ = Phase

2. M7032/1-1 and /1-2 inverters shall be provided with a time totalizing meter conforming to MIL-M-7793/6. The meter shall be located on the left side of the control box facing the connector end.

3. MIL-PRF-7032 variance: The inverters shall comply with MIL-PRF-7032 except that the following paragraphs of MIL-PRF-7032 shall be modified:

3.4.2.1.1 External electric connections. Delete the last sentence and substitute: "Input and output connections for appropriate single-phase and three-phase operation shall be as shown on figure 3 of this specification sheet.

3.4.2.1.1.3 Output voltage jacks. Delete the first sentence and substitute: "One set of placarded pin jacks shall be provided on the inverter as shown on figures 1 and 2 of this specification sheet, between which the output voltage can be measured."

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:48Z
Check the source to verify that this is the current version before use.

3.4.2.2.2.1 Voltage setting. At the end of the paragraph: Add: "For the M7032/1-1 and M7032/1-2 inverters, a means of voltage adjustment shall be located on the connector end above the horizontal centerline of the inverter."

3.4.3.5 Cooling. Delete the last sentence and substitute the following: "The direction of air flow shall be as follows:"

M7032/1-1 Airflow exhaust from center
M7032/1-2 Airflow exhaust from connector end
M7032/1-3 Airflow exhaust from the end opposite the connectors
M7032/1-4 Airflow exhaust from the connector end
M7032/1-5 Airflow exhaust from the connector end
M7032/1-6 Airflow exhaust from the end opposite the connectors

3.4.3.8 Size and weight. Delete the first sentence and substitute: "The size and dimensions of the inverters shall be as shown on figures 1 and 2 of this specification sheet. Unless otherwise specified, all dimensions are in inches and tolerances are ±0.015 inch. Inverter weights shall be as required in table I of this specification sheet."

3.5.1.1.a Delete the subparagraph in its entirety and substitute: "The normal input voltage range shall be as required in table II of this specification sheet."

3.5.1.1.b Delete the subparagraph in its entirety and substitute: "The abnormal input voltage range shall be as required in table II of this specification sheet."

3.5.1.1.c Delete the subparagraph in its entirety and substitute: "The low input voltage range shall be as required in table II of this specification sheet and applies only to M7032/1-5 inverter.

3.5.1.3 Output frequency regulation. Delete the second sentence and substitute: "The steady state frequency for rotary inverters, M7032/1-1 and M7032/1-2, shall be as required in table II of this specification sheet. Static inverters shall maintain a steady-state frequency of 400 ± 7 Hertz under all specified input voltages (see 3.5.1.1), load, temperature, and altitude conditions specified (see 4.5.2)."

3.5.1.4 Efficiency. Delete the first sentence and substitute: "The minimum efficiency of rotary inverters shall be 50 percent for 3-phase and 45 percent for 1-phase inverters (see 4.5.3). "

3.5.1.7.1 Static inverters. Delete the first sentence and substitute: "The steady-state output voltage limits within two seconds after application of input voltage for conditions 1 through 5 shall be as required in table II of this specification sheet. For the M7032/1-6 inverter only, the output voltage during input power application shall be within and remain within limits of this specification sheet within 400 milliseconds after application of input voltage."

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:48Z
Check the source to verify that this is the current version before use.

3.5.1.7.2 Rotary inverters. Add the following new second and third sentences: "The maximum controlled output voltage is condition 6, as required in table II of this specification sheet. The steady-state output voltage limits for conditions 1 through 5 shall be as required in table II of this specification sheet."

3.5.1.8.1 Overload. Add the following sentence at the end of the paragraph: "The minimum output voltage during overload conditions is condition 7, as required in table II of this specification sheet."

3.5.1.10.1 Voltage. Add the following sentence to the end of the paragraph: For M7032/1-6 only, the output voltage modulation shall not exceed 1.0 percent under normal input voltage conditions at full load, half load, and no load, and 1.5 percent from 24.0 to 26.0 volts DC (see 3.5.1.1 and 4.5.9.1).

3.5.1.16 Acoustical noise. Not applicable.

3.5.2.1 Temperature altitude. Delete the paragraph in its entirety and substitute: "All inverters covered by this specification sheet are class B. The inverter shall be capable of operation without degradation of performance throughout the temperature-altitude range specified in 3.5.2.1.2."

3.7.1 Terminals. Delete the paragraph in its entirety and substitute: "The input and output connectors and voltage pin jacks shall be marked as shown on figures 1 and 2 of this specification sheet (see 4.5.1)."

3.7.4 Airflow. Delete first sentence of the paragraph and substitute: When an integral cooling fan is used, direction of airflow shall be plainly and permanently marked on the outside of the housing in the direction specified in 3.4.3.5 of this specification sheet."

VERIFICATION:

4.2.2 Qualification inspections and tests. Add the following new second sentence: "There are no additional examinations and tests required by this specification sheet. Testing shall be in accordance with MIL-PRF-7032 as modified by this specification sheet."

4.2.3 Qualification retention. Add the following new second sentence: "There are no additional qualification retention examinations and tests required by this specification sheet. Testing shall be in accordance with MIL-PRF-7032 as modified by this specification sheet."

4.4.1 Test conditions. Delete the paragraph in its entirety and substitute: "Unless otherwise specified in MIL-PRF-7032 or in MIL-STD-810, tests shall be conducted under the following conditions:."

4.4.1.6 Warm-up and stabilization. Add the following new second sentence: "Inverter full rated load values shall be as required in table 1 of this specification sheet."

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:48Z
Check the source to verify that this is the current version before use.

4.5.1 Visual and mechanical inspection. Add the following sentences to the end of the paragraph: “Inverter weight shall be as required in table 1 of this specification sheet. Inverter dimensions and markings shall be as shown on figures 1 and 2 of this specification sheet.”

4.5.2.2 Power factors. Delete the first sentence and substitute: “The tests specified in 4.5.2.1 shall be repeated at the minimum lagging and minimum leading power factors in table I of this specification sheet.”

4.5.10.1.1.b Delete the subparagraph in its entirety and substitute: Under the conditions specified in 4.5.2.1.a, b, and c with steady-state minimum lagging power factor shall be as required in table I of this specification sheet.”

4.6.1 Temperature cycling. Delete the second sentence and substitute: Twenty-five of the operating hours shall be at unity power factor, the remaining 25 hours at the minimum lagging power factor required in table I of this specification sheet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:48Z
Check the source to verify that this is the current version before use.

TABLE II. Operating conditions.

For any combination of conditions listed within each section below, the output voltage and frequency shall remain within the indicated limits.	OUTPUT LIMITS (LIMITS APPLY TO EACH PHASE FOR 3-PHASE OPERATION)					
	1/ M70321/1-1, /1-2		1/ M7032/1-3, /1-4, /1-6		M7032/1-5	
	RMS ∅ VOLTAGE	FREQ (HZ)	RMS ∅ VOLTAGE	FREQ (HZ)	RMS ∅ VOLTAGE	FREQ (HZ)
A. NORMAL INPUT VOLTAGE 1. Input voltage varied between 22.0 and 29.0 volts. 2. Load varied from zero to full load. 3. Ambient temperature-altitude varied per Class B. 4. Load power factor varied from .75 lag or .80 lag (as applicable) to .95 lead 5. Unit mounted in any position.	 Low 112.5 High 117.5	 Low 390 High 410	 Low 112.5 High 117.5	 Low 393 High 407	 Low 112.5 High 117.5	 Low 393 High 407
B. ABNORMAL INPUT VOLTAGE 6. Variations 2, 3, 4, and 5 above with input voltage varied between 20.0 and 31.5 volts.	Low 100 2/ High 120 2/	Low 350 2/ High 410 2/	Low 108 High 117.5	Low 393 High 407	Low 108 High 117.5	Low 393 High 407
C. OVERLOAD 7. Variations 1 through 5 above except load varied from rated full load current to 200% rated full load current for 5 seconds.	Low 50 High 117.5	Low 385 High 410	Low 50 High 117.5	Low 393 High 407	Low 50 High 117.5	Low 393 High 407
D. LOW INPUT VOLTAGE 8. Input voltage varied between 12 and 20 volts for 3 minutes. 9. Load impedance varied from infinity to 26.5 ohms for the M7032/1-4 and /1-5 inverters, and from infinity to 106 ohms/phase for the M7032/1-3 and /1-6 inverters. (b) 10. Ambient temperature varied between -35 °C and +40 °C at sea level with inverters fully warmed up, and between -35 °C and +85 ºC at sea level with inverters not previously warmed up. 11. Variations 4 and 5 above.	 No Operation Required	 No Operation Required	 Low 45 High 117.5	 Low 393 High 407	 Low 108 High 117.5	 Low 393 High 407

1/ The M7032/1–1 and –6 inverters shall operate either in the wye or delta mode when external connections are made as shown on figure 3.
2/ For this condition, the line-neutral frequency/voltage ratio shall remain between 3.0 and 4.0.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:48Z
Check the source to verify that this is the current version before use.

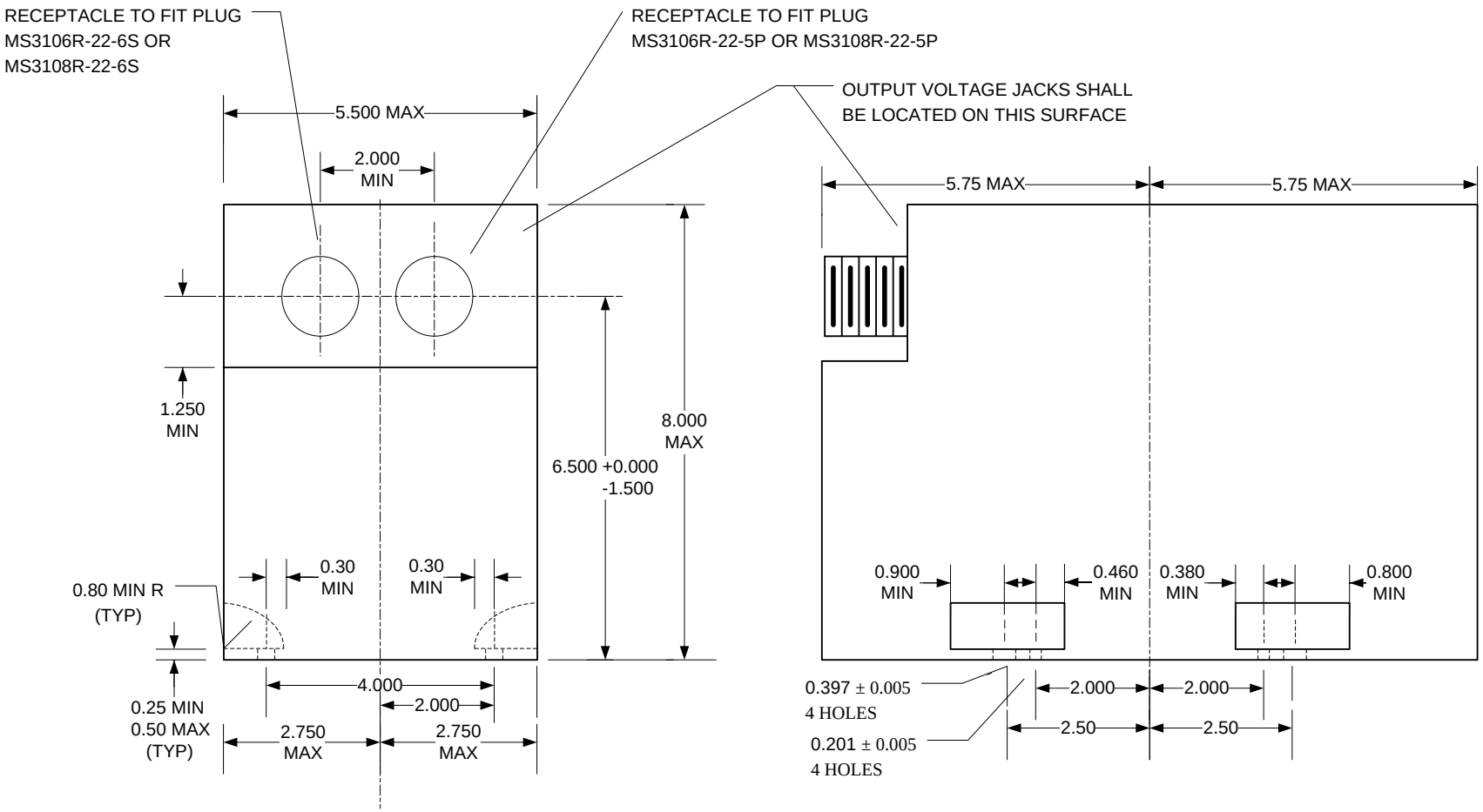


FIGURE 1. Rotary inverter (M7032/1-1, M7032/1-2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:48Z
Check the source to verify that this is the current version before use.

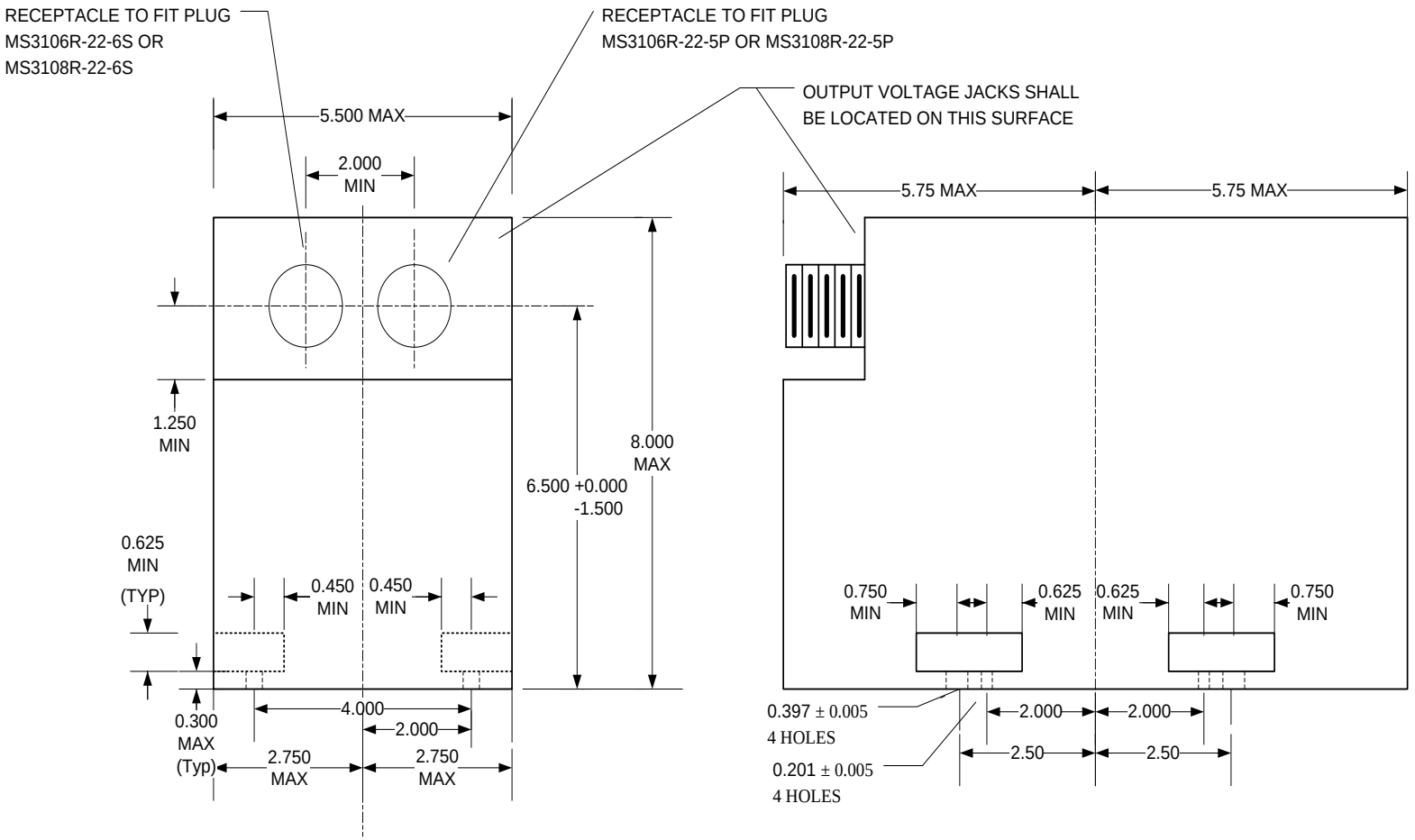


FIGURE 2. Static inverter (M7032/1-3, M7032/1-4, M7032/1-5, M7032/1-6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:48Z
Check the source to verify that this is the current version before use.

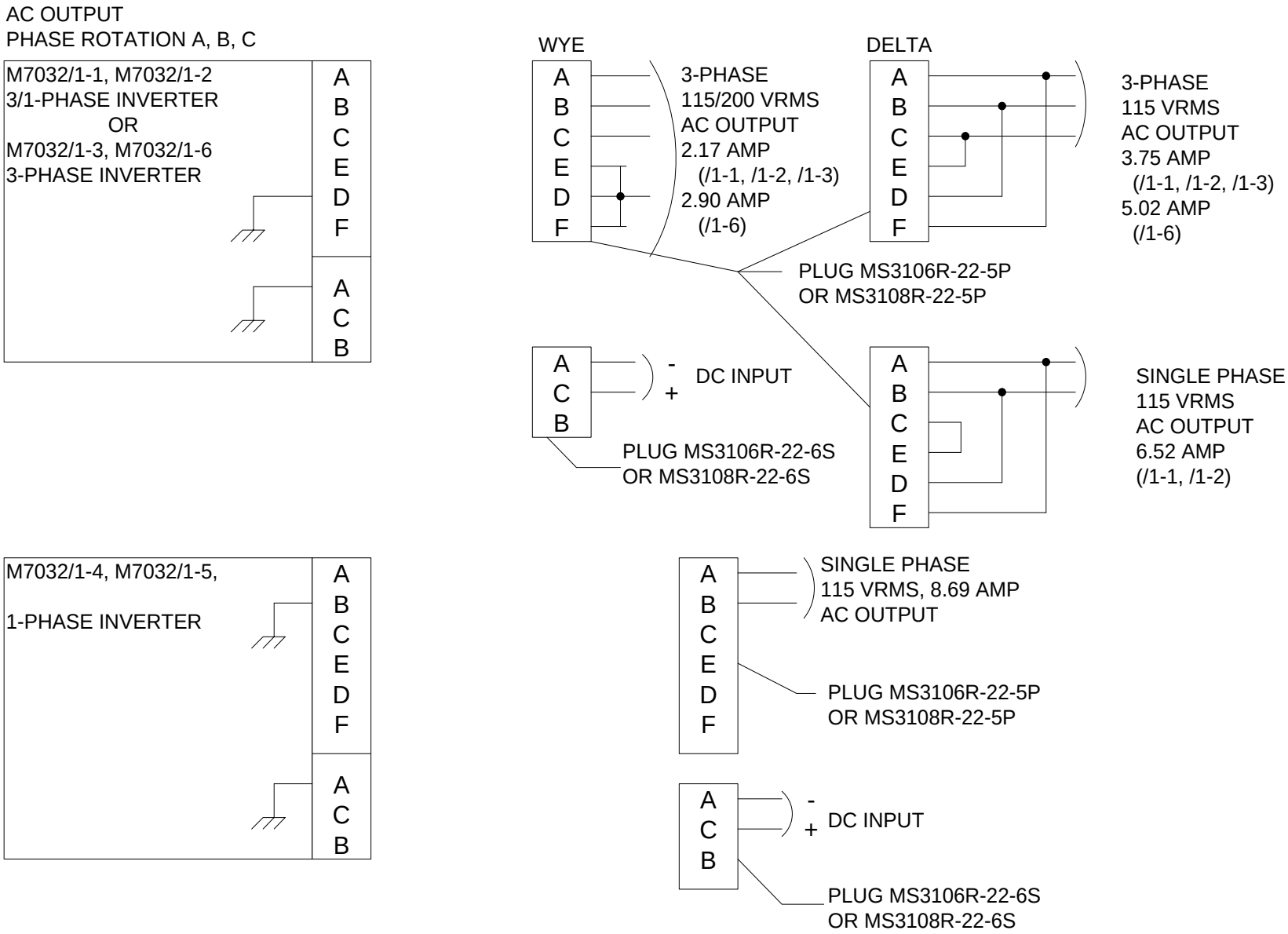


FIGURE 3. External connections diagram.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:48Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army – CR4
Navy – AS
Air Force – 99

Preparing activity:
Navy - AS

(Project 6125-2010-001)

Review activities:
Army – AV, CR
Air Force – 71
DLA – GS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:48Z
Check the source to verify that this is the current version before use.